Data Analysis and Machine Learning with PYTHON

A selection of the best DataCamp courses (and other items)

Gilles MOREL – UTC – September 2023

To choose the best courses fitted to your level and needs :

- Go the entry page (clic on the title link below) to read the description and program
- See the prerequisite before beginning

Python the language (basics and advanced)

For those who don't have the basics of the Python language, before using ML libraries

- **Introduction to Python** (includes introduction to NumPy)
- **Intermediate Python** (includes introduction to Matplotlib and Pandas)
- Python Data Science Toolbox (Part 1)
- Python Data Science Toolbox (Part 2)

For those who want to go further in Python (special topics)

- Writing functions in Python
- Object-oriented Programming in Python
- Introduction to Databases in Python
- Introduction to importing data in Python
- Working with Dates and Times in Python
- Software Engineering for Data Scientists in Python

Test your Python skills with : Python programming skill assessment

Data Manipulation and Pre-Processing (preference with pandas better than NumPy)

The basics :

- **Data Manipulation with pandas**
- Cleaning data in Python

To go further

- Joining Data with pandas
- Streamline Data Ingestion with pandas
- Reshaping Data with pandas
- Writing Efficient Code with pandas

Test your skills about data manipulation with Python

Data Visualization

The basics :

- Introduction to Data Visualization with Matplotlib
- Introduction to Data Visualization with Seaborn
- Introduction to Data Visualization with Plotly in Python
- Visualizing Time Series Data in Python

Test your skills about Data visualization with Python

To go futher

- Intermediate Data Visualization with Seaborn
- Improving your Data Visualization with Python
- Building Dashboards with Dash and Plotly
- Visualizing Geospatial Data with Python (using GeoPandas and Folium)

EDA (Exploratory Data Analysis) and statistics

- Exploratory Data Analysis (EDA) with Python
- Statistical Thinking in Python – Part 1
- Statistical Thinking in Python – Part 2

Test your skill in statistics fundamentals with Python

Machine Learning (and IoT)

The basics :

- Supervised Learning with scikit-learn
- Unsupervised Learning in Python

Test your skills about machine learning fundamentals with Python

IoT + machine learning, complete process :

- Analysing IoT Data in Python

To go further :

- Linear Classifiers in Python
- Machine Learning with Tree-Based Models in Python
- Cluster Analysis in Python

Time Series Analysis

- Manipulating Time Series Data in Python
- Time Series Analysis in Python
- Visualizing Time Series Data in Python
- Machine Learning for Time Series Data in Python
- ARIMA Models in Python

Introduction to Deep Learning

The basics

- Introduction to Deep Learning in Python
- Introduction to Deep Learning with Keras

To go further

- All the courses Deep Learning with Python tools (DataCamp track)

Projects and use cases with DataCamp and Jupyter

- Introduction to DataCamp projects with Python and Jupyter
- DataCamp available projects in Python
- DataCamp case studies in Python
- DataCamp competitions